Lemma: G plane triangulation with $v(G) \geq 4 \Rightarrow G$ is 3-connected.

proof:

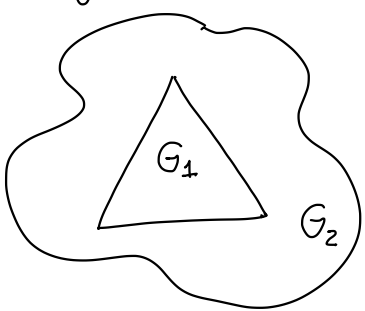
Proposition: H plane triangulation, $v(H) = n \geq 4 \Rightarrow \exists \geq 6$ edges $e \in E(H)$ s.t. the only Δ's containing e are faces.

Call them good edges

proof:

Induction on n. B.I: K_4

Let T be a triangle that is not a face.



G_1 = induced graph on T and all vtxs inside.

G_2 = ———————— outside.

By induction: G_1 has $v(G_1) \geq 6$ good edges (w.r.t. G_1)

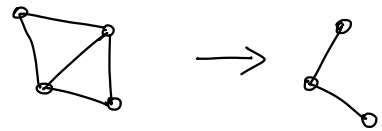
G_2 has $v(G_2) \geq 6$ ——— (w.r.t. G_2).

Since no edge crosses the boundary of T,

e is good edge w.r.t. $G_i - E(T) \Rightarrow e$ is good edge w.r.t. G.

$\#$ good edges in $G \geq 6 + 6 - 6$ $\blacksquare$

<u>Obs</u> : e good edge $\Rightarrow$ G/e is maximal planar



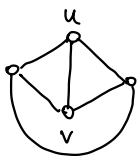
(as we lose only 3 edges)

Suppose for contradiction that G is not 3-connected.

Then, $\exists\ u, v \in V(G)$ s.t. $G - u - v$ is disconnected.

<u>Obs</u> : Either $uv \notin E(G)$ or uv is not good.

$\Rightarrow$ After $n-4$ edge contractions along good edges, we arrive at K_4 ! (Max planar on 4 vtxs)

$\Rightarrow$ 

But $G-u-v$ does not disconnect the graph! Contradiction $\blacksquare$

<u>Fáry's thm</u>: Let G be a maximal planar graph and $K \subseteq G$ so that $K \cong K_3$. Then, G can be drawn in the plane without crossings so that its edges are straight line segments and the outerface F is so that $G[F] = K$.

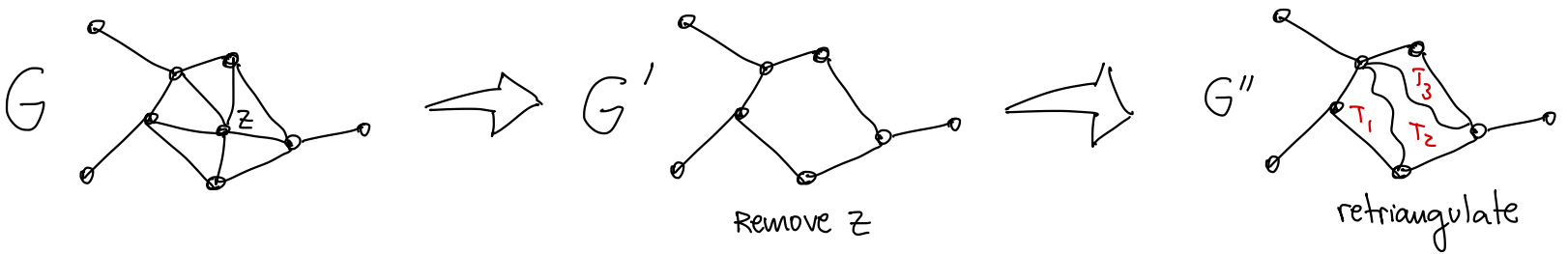
<u>proof</u>: Let $K \subseteq G$ be a triangle an call $V(K) = \{a, b, c\}$.

Induction on $n = v(G)$. B.I.: $n = 4$

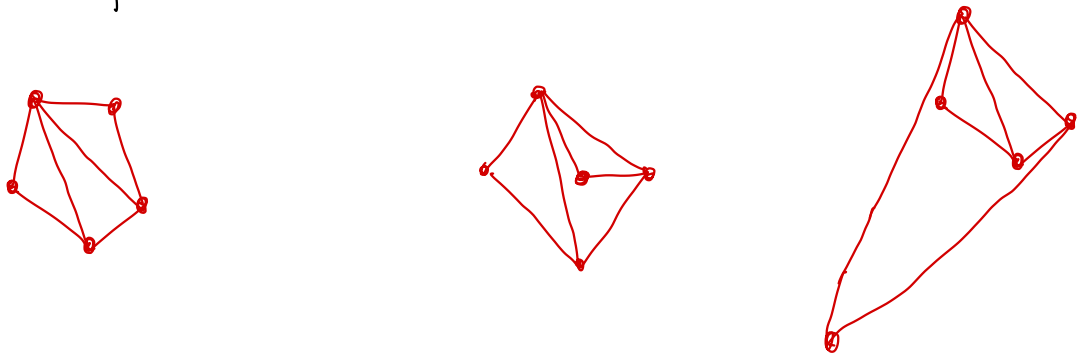
Deficiency of v: $r(v) := 6 - d_G(v) \leq 3$ (As G is 3-connected)

$$\Rightarrow \sum_{v \in V(G)} r(v) = 6n - 2 \cdot (3n - 6) = 12$$

$\Rightarrow \exists \geq 4$ vtxs v st $r(v) \geq 1$. $\Rightarrow \exists\, z \neq a, b, c$ st $d_G(z) \leq 5$.



By induction, G'' can be drawn in the plane without crossings so that its edges are straight and the outerface F is so that $G[F] = K$.

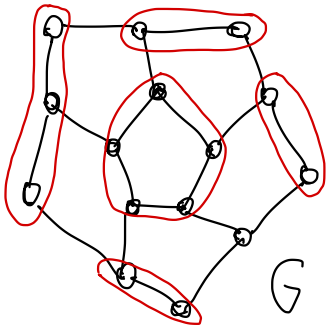


Look at G' in the plane embedding of G''. Remove $E(G'') \setminus E(G')$ and add Z back using the <u>art gallery thm</u>.

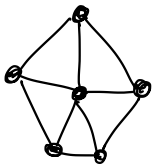
<u>Thm</u>: In an art gallery shaped as a 5-sided polygon one needs one guard to observe the whole gallery.

Defn: A graph X is a <u>minor</u> of a graph G if X can be obtained from G by a series of edge contractions, edge deletions and vertex deletions.

Example:



G



X

X is a minor of G.

Lemma: X is a minor of $G \iff \exists$ collection of disjoint vtx sets $(V_x)_{x \in V(X)}$ in $V(G)$ such that

① $xy \in E(X) \Rightarrow e(G[V_x, V_y]) \geq 1$.

② $G[V_x]$ is connected $\forall\, x \in V(X)$.